

RateWalk: Forecasting Fed Rate Decisions with a Shrunk Markov Chain

An out-of-sample, point-in-time study of whether inflation regimes help

Tanishk Yadav

MS Financial Engineering, New York University

tanishkyadav@nyu.edu

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Abstract

I ask whether a memoryless Markov chain on the Federal Reserve’s policy moves can forecast the next rate decision out of sample, and whether conditioning the chain on the inflation regime improves it. Under strict no-look-ahead discipline, with CPI read at its real ALFRED release date so revisions never leak, the answer is a clean bias-variance story. A first-order chain beats a climatology baseline. *Naive* conditioning on the CPI regime overfits and *hurts*, because it splits sparse transition data across regimes. Shrinking each regime row toward the pooled chain (an empirical-Bayes Dirichlet prior) recovers a small but consistent edge that *replicates across the United States, the United Kingdom, and Germany*, survives a data-driven shrinkage strength, and sits on a broad parameter plateau rather than a tuned point. Against the market’s own pricing, proxied by a walk-forward-calibrated Treasury-curve signal, the chain wins overall (log-loss 1.06 vs. 1.20) but the decomposition is sharper: the market is better on 82% of actual move months, the chain on 85% of holds, and a no-look-ahead adaptive blend beats both. The edge lives in probabilistic calibration: a duration-timing strategy built on the signal does not beat a constant-duration benchmark risk-adjusted. I report the negatives as part of the result.

1 Motivation

Most discretized rate models are evaluated in sample, where adding parameters always looks helpful, and many quietly use revised macro data unavailable at decision time. I impose two disciplines from the outset. First, **no look-ahead**: every prediction at month t is built only from data public before t . Second, **point-in-time CPI**: I use the initial-release value dated by its real publication date (FRED/ALFRED, `output_type=4`); the initial-release year-over-year figure differs from the later revised series in roughly four of five months, by up to a few tenths of a point (Figure 1).

2 Data and states

I use the daily effective policy rate, the curve, and CPI from FRED for the US, UK, and Germany, 1990-2026. The monthly rate change is snapped to the FOMC move grid $\{-50, -25, 0, +25, +50, +75\}$ bps; CPI year-over-year is binned into five inflation regimes. A first-order chain is the matrix $P(\text{next move} \mid \text{current move})$; the conditional chain adds the regime, $P(\text{next} \mid \text{current, regime})$. Rates are sticky: the no-change state dominates every row (Figure 1, right inset of the transition structure in the companion notebook).

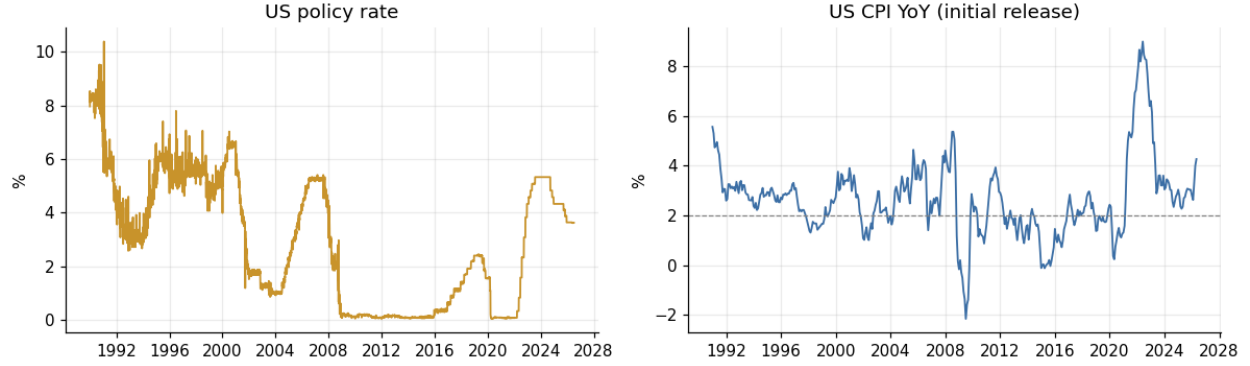


Figure 1: US policy rate and initial-release CPI year-over-year, 1990-2026. Using the revised CPI instead of the initial release would leak information unavailable at decision time.

3 The out-of-sample test

At each month I refit the chain on past-only data and predict the distribution of the next move, scoring mean log-loss (lower is better) against a climatology baseline (the unconditional marginal of moves) and the unconditional chain. The first-order chain beats climatology (1.071 vs. 1.162 for the US), confirming genuine predictive content. But *raw* CPI conditioning is *worse* (1.136): with the non-hold transitions split across five regimes, each regime-row is estimated from a handful of observations and the variance swamps any signal.

4 The fix: shrinkage

If the failure is variance, the remedy is shrinkage. I place a Dirichlet prior on each regime row centered on the pooled (unconditional) row with strength τ pseudo-counts,

$$\hat{p}(j \mid i, r) = \frac{c_{r,i,j} + \tau \bar{p}(j \mid i)}{n_{r,i} + \tau},$$

so that $\tau \rightarrow 0$ recovers the raw conditional model and $\tau \rightarrow \infty$ the unconditional one. A sweep (Figure 2) shows a broad interior plateau, $\tau \approx 20$ to 500, over which the shrunk conditional model *beats the unconditional chain*. The macro signal was real; it was drowned in variance, and regularization recovered it.

5 Robustness

Replication. The pattern (raw conditioning hurts, shrinkage helps) holds in all three sovereigns (Table 1, Figure 3). **No tuned τ .** A fully data-driven empirical-Bayes τ , estimated per-step from past-only data by maximizing the Dirichlet-multinomial marginal likelihood, also beats the unconditional chain in all three with no tuning; it is marginally worse than a fixed moderate τ only because it is conservative when early data is thin, then adapts (settling near 130 for the US, 300 for the UK, and 17 for Germany; German decisions are the most regime-dependent). **Cadence is not the lever.** Re-running at the Fed’s roughly eight-meeting annual cadence rather than monthly does not systematically help; even at meeting frequency 60 to 80% of decisions are holds, because policy is intrinsically sticky.

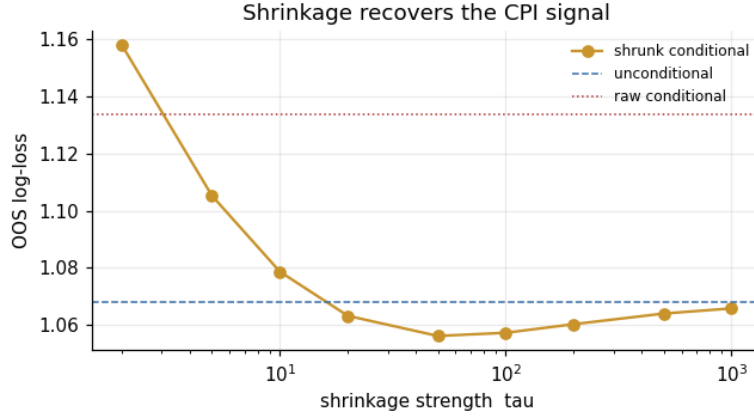


Figure 2: Out-of-sample log-loss of the shrunk conditional model versus shrinkage strength τ (US). The shrunk model beats the unconditional chain over a broad plateau, not a tuned point.

Model (out-of-sample mean log-loss)	US	UK	Germany
climatology	1.162	1.010	0.615
unconditional chain	1.071	0.843	0.596
conditional, raw	1.136	0.904	0.662
conditional, shrunk ($\tau=50$)	1.059	0.819	0.549
conditional, empirical-Bayes τ	1.065	0.843	0.563

Table 1: Out-of-sample mean log-loss (lower is better). Raw conditioning is worse than the unconditional chain everywhere; the shrunk and empirical-Bayes versions are better everywhere.

6 From a forecast to a strategy

A calibrated forecast is not yet money. Turning the model’s expected move into a duration tilt (expect cuts, lengthen duration; expect hikes, shorten) and pricing the realized return off the actual Treasury curve, the timing strategy does *not* beat a constant two-year bond risk-adjusted (Sharpe ≈ 1.06 vs. 1.39; Figure 4). It tilts long-duration only about 7% of the time, and those tilts add volatility without enough return. The forecasting edge is real but lives in calibration, not in this simple tradeable rule.

7 Does it beat the market?

The question any rates desk asks of a policy model. Fed funds futures are the canonical market forecast but are not freely available historically, so I build the standard free proxy from the short end of the Treasury curve: the spread between the 3-month bill yield and the current policy rate carries the market’s directional view of upcoming policy. To avoid hardcoding how many basis points of next-month move a given spread implies, the mapping is itself calibrated walk-forward: at each date I regress realized next-period moves on past spreads (past pairs only), predict the mean move from the current spread, estimate the residual scale from past errors only, and integrate a normal over the move grid. The market proxy is then scored with exactly the same out-of-sample log-loss, over exactly the same 319 monthly decisions, as the Markov models.

Three results (Figure 5, top). First, the shrunk chain beats the curve proxy overall: mean log-loss 1.063 versus 1.199. Second, and more interesting, the two contain *complementary* information

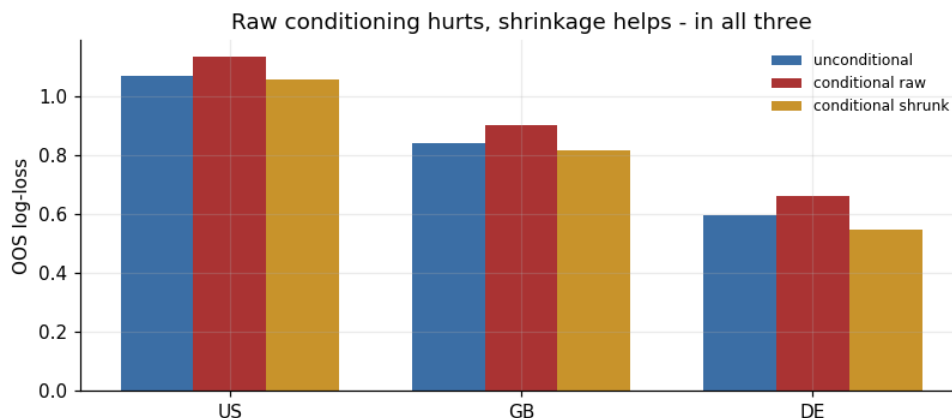


Figure 3: The flip replicates: raw conditioning hurts and shrinkage helps in the US, UK, and Germany.

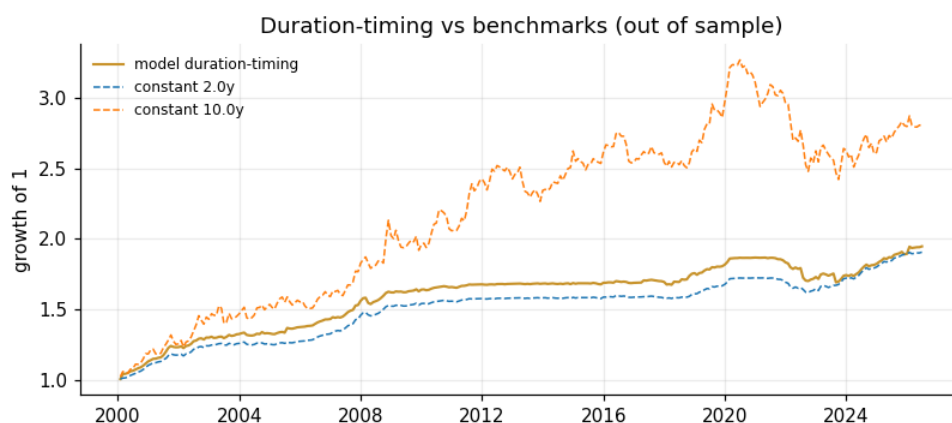


Figure 4: Out-of-sample growth of \$1: the model’s duration-timing strategy versus constant-duration benchmarks. The strategy does not beat the constant two-year bond risk-adjusted.

(Figure 5, bottom left): on the 83 months where the Fed actually moved, the market proxy is the better forecaster 82% of the time, while on the 236 hold months the chain is better 85% of the time. The cumulative-loss path shows the same structure in time: the curve proxy led through the 2000s hiking and cutting cycles, and the chain’s advantage is earned almost entirely in the post-2009 hold-dominated era. The chain knows that policy is sticky; the curve knows when something is coming.

Third, the natural consequence: a parameter-free combination beats both. I pool the two forecasts linearly with weights set, at each date, by the components’ inverse mean log-loss on already-realized decisions only (no tuned constant, no look-ahead), giving mean log-loss 1.056 and top-choice accuracy 72.7% versus 68.7% for the chain alone.

Honesty notes. The proxy is a curve-implied approximation of the futures strip, handicapped by a single linear calibration; the right reading is “the chain beats a reasonable free proxy of market pricing and complements it,” not “the chain beats the market.” A test against actual fed funds futures is the natural extension, and the moves/holds decomposition predicts what it would show: the market’s edge concentrates exactly where positioning matters most.

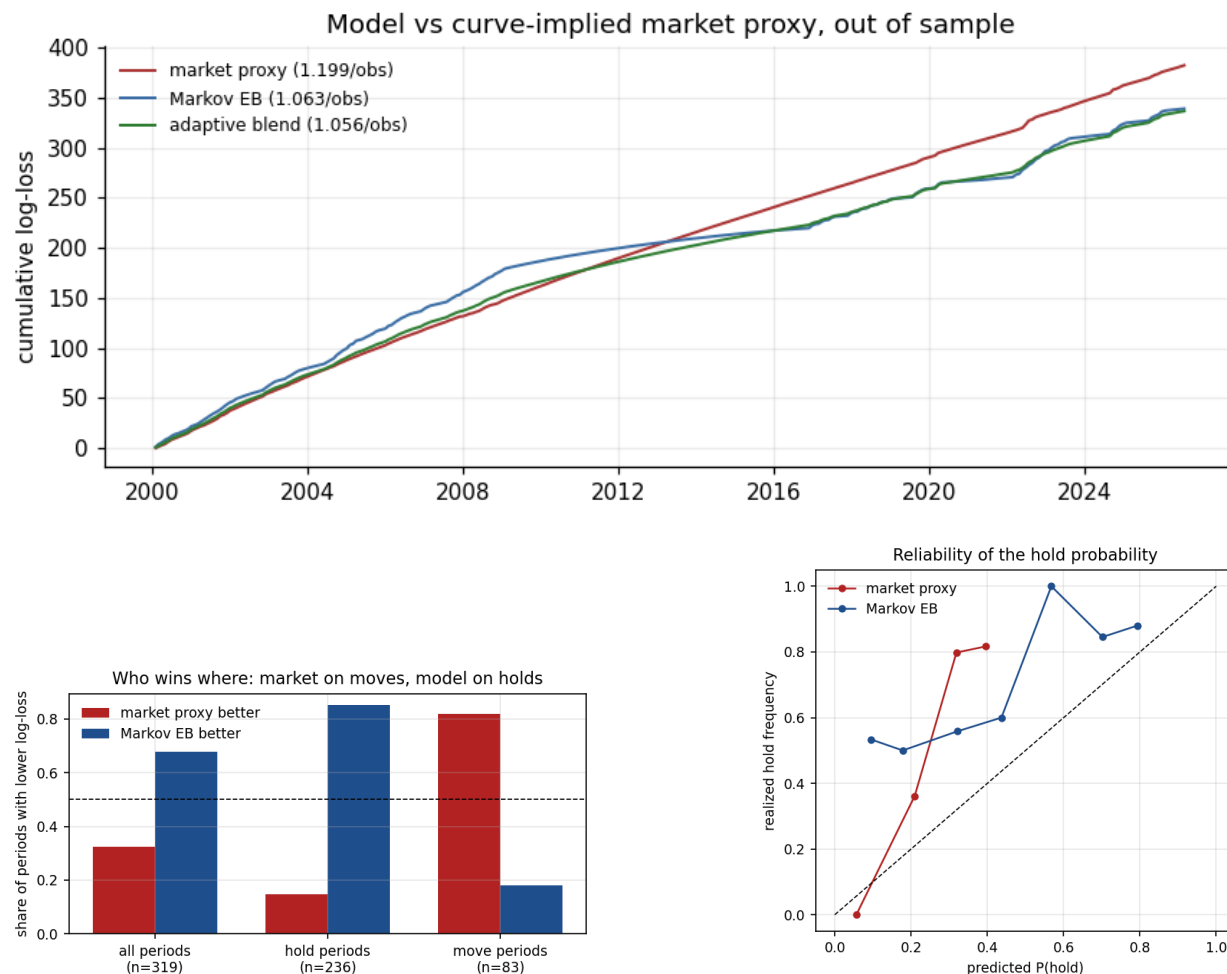


Figure 5: Model versus curve-implied market proxy, out of sample (US, 319 monthly decisions). Top: cumulative log-loss; the proxy led through the 2000s cycle era, the chain wins the post-2009 hold era, and the adaptive blend tracks the better of the two. Bottom left: the market is the better forecaster on move months, the chain on hold months. Bottom right: both $P(\text{hold})$ forecasts are roughly calibrated, with the chain tighter at high confidence.

8 Conclusion

A memoryless Markov chain on Fed moves beats climatology out of sample; naive inflation conditioning overfits; empirical-Bayes shrinkage recovers a small, consistent, cross-country edge; cadence is not the lever; the chain beats a curve-implied proxy of market pricing overall while the proxy wins precisely on move months, and a no-look-ahead adaptive blend of the two beats both; and the edge is in calibration rather than a tradeable strategy. Every result is reproducible from a seed under strict no-look-ahead with point-in-time CPI, and the honest negatives are part of the finding. Natural extensions are the exact published FOMC calendar (to settle the cadence question), actual fed funds futures as the market benchmark, and a richer mapping from forecast to position.